

ally associate with the cosmetics industry, including Fujifilm and BASF. Of 367 unique inventions filed in 2009, 10 were by Fujifilm; 9 were by BASF and 7 were by Amorepacific.

3. **“Nano” Trademarks:** From 2000 through the end of 2009, a total of 217 personal-care brands that incorporate the term “nano” were trademarked in the United States, United Kingdom, Canada, European Community and WIPO. In 2009, 124 patents had been filed with WIPO, more than double the 51 patents filed in the United States. China and Korea round out the top three, following closely behind with 50 patents each.

The report predicts that the popularity and use of nanotechnology in personal-care products is certain to expand in the coming years, especially as consumers begin to experience the benefits of this technology and associate it with looking younger and feeling rejuvenated.

However, the dream conjured up by Thomson Reuters has also run into criticism, as just being an advertisement of their IP Market Reports, and failing to mention potential health and safety risks that are being raised by scientists. So let’s break down the mystery of how exactly beauty products make use of nanotechnology and what are the potential risks involved.

How does it work?

The epidermis (top layer of skin) on the face is about 0.2 millis thick. Although it seems like a minuscule quantity, we need to look at this dimension from the point of view of the ingredients in the skin care product. As far as they are concerned, it may as well be a mile thick. Many products on the market are considered “scientifically unlikely to make significant improvements” because they don’t penetrate the epidermis; they simply create a nutrient rich barrier between the environment and the skin.

Nanotechnology has enabled cosmetic producers to formulate ingredients with particles that are 100,000 times smaller than the width of a human hair. Nanotechnology makes it possible to control the molecules and therefore active ingredients are small enough to penetrate the skin. These invasive technologies can tear at the skin or can be injected under the skin. The surface of the skin has dead skin cells but also a lipid (fat) layer that contains very, very small gaps. These gaps are so small that nothing could get between them, but that is of course, until nanotechnology came along. Nanotechnology allows scientists to develop skin care products that deliver active ingredients right to the heart of the skin.

Also these minute nanoparticles do not lump together but disperse uniformly in a flat layer, enabling light to pass through them. This technology has already